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AI Papers on the Spectrum of RSI and the Illusion of Thinking

"To be or not to be", is also the question for AI.

MICHAEL SPENCER AND TOBIAS MARK JENSEN
JUL 10



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Apple's Illusion of Thinking Paper, the RSI papers and the academics of public relations around Generative AI.

In recent times it appears academic papers are being used as a form of public relations for various ideological positions along the AI hype cycle spectrum. This isn't really anything new but it is striking.

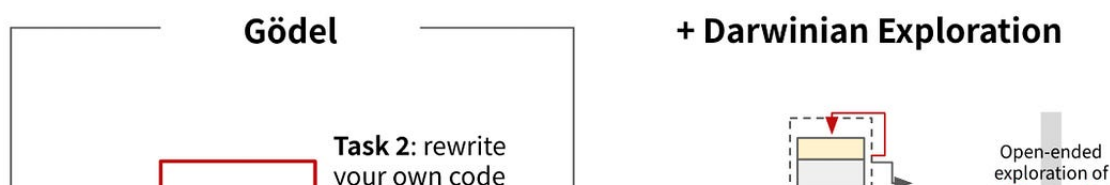
Recursive Self-improving AI (RSI)

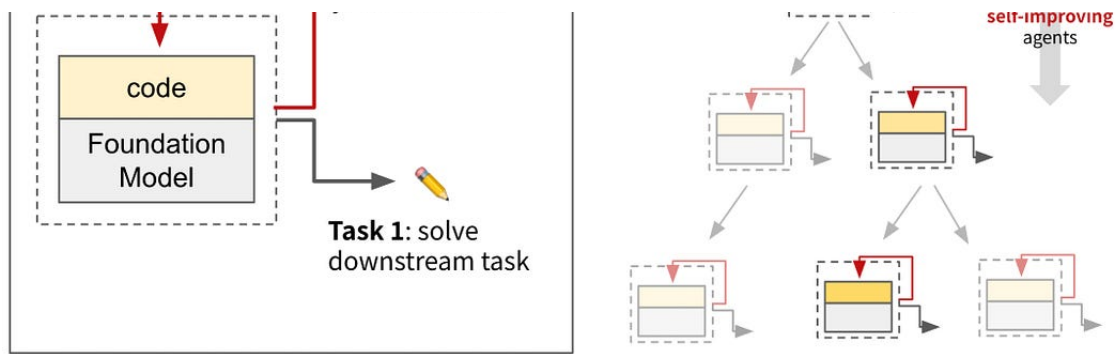
A couple of papers caught my attention around this [trend of RSI](#), when in mid May Google DeepMind [announced AlphaEvolve](#). This also comes across in Anthropic claiming how much Claude Code is used to make Claude Code itself better, i.e. claims like [Claude Code wrote 80% of its own code](#).

[Read the AlphaEvolve Paper](#)

While Google says its deployed algorithms discovered by AlphaEvolve across Google's computing ecosystem, including our data centers, hardware and software. There are also other research labs claiming other kinds of aspects of RSI or frameworks for it. While far fetched by the usual audacious claims by DeepMind, Sam Altman's [Gentle Singularity](#) blog is even more extreme. As if the prophecy of Venture Capitalists could justify the enormous capital being spent on a pattern matching system like the transformer architecture where token prediction are coached or imbued with names like "reasoning models" and "superhuman intelligence".

The appeal of [Meta learning](#) and a [Gödel Machine](#) of a hypothetical self-improving AI system that optimally solves problems by recursively rewriting its own code when it can mathematically prove a better strategy doubles down on the illusion of thinking and morbid anthropophication that AI can magically arrive itself at some synthetic ultimate synesthetic form of self-awareness in AI, like an agent that can reason about its own limitations and modify itself accordingly.





LLMs as Utopia Bringers

The latest to reincarnate the Gödel machine is indeed [of course Sakan AI](#). Where [David Ha](#) worked at Google for a long time and is familiar with such academic dreams. In the age of commercial AGI marketing, somehow the transformer architecture is elevated [into mythic proportions](#) where AI machines reach almost poetic connotations of utopian outcomes as depicted by the CEO of Anthropic in his [Machines of Loving grace](#) blog.

[Read the Sakana AI Paper](#)

Which brings us to the papers by Apple around the capabilities of LLMs, and of course ammunition for the Gary Marcus types of this world. The latest of which has enabled [Marcus on AI](#) to [fan of flames](#) of righteous skepticism. Apple who have always been somewhat pragmatic about the limitations of the transformer architecture and not follow BigTech in its AI infrastructure capex frenzy chasing fantasies of AGI machines.

The Transformer architecture from [a 2017 Google paper](#) is just one possible architecture that will contribute to improved AI capabilities. Calling pattern matching systems “reasoning models” certainly doesn’t feel sincere and neither conflating their capabilities with human PhDs seem ethical. The **Anthropomorphic hacks** of human psychology have been taking to exceptional dimensions with Sam Altman’s style of leadership, even to the extent that AI influencers have downplayed and demeaned the Apple paper.

[Read the Paper](#)

That [Apple published the paper](#) mere days before WWDC that would go on to underwhelm with Generative AI announcements was one of the most hilarious and symbolic things in the state of AI in 2025. The paper was titled *The Illusion of Thinking: Understanding the Strengths and Limitations of Reasoning Models via the Lens of Problem Complexity*. Critics [have appeared](#) on all sides trying to debunk [or dismiss the paper](#).

I asked [Tobias Mark Jensen](#) of [Futuristic Lawyer](#) for this take, which is the deep dive of today below.

[Sakana AI](#) claim that experiments demonstrate the Darwin Gödel Machine can continuously self-improve by modifying its own codebase. So now we do not just just trendy acronyms slike AGI (artificial general intelligence), we have:

- ASI: artificial super-intelligence.
- RSI: recursive self-improving AI.
- DGM: Darwin Gödel Machines
- Unfortunately the Generative AI skeptics have not provided their own acronyms for their pet ideas.

Comparison with AlphaEvolve

While both systems adopt a similar evolutionary framework, their scopes and methodologies differ in the following ways:

Feature	AlphaEvolve	DGM
Focus	Evolving functions and codebases	Evolving the agent itself
Level of Innovation	Algorithmic level	Agent-level (toolset, methodology)
Role of LLM	LLM acts as “genetic operators” to modify algorithms	LLM serves as the “brain” to evolve itself with better tools and strategies
Evaluation	Well-defined problems with automated evaluators	Open-ended environments

Nor is this the first time that Apple have done research that seems to downplay the capabilities of LLMs and the transformer architecture framework. Meanwhile Apple even with WWDC 2025 not delving into Gen AI have actually announced [on-device Foundation](#) Language models.

Meanwhile both Google and [Apple finally face](#) some rule of law over their monopolistic gate keep activities. Google may have to [divest chrome and stop paying partners](#) to put Google first. Namely that's [\\$20 billion in lost revenue](#) likely for Apple.



What Apple is Saying in the Illusion of Thinking Paper

Apple is basically saying that AI reasoning isn't all it's cracked up to be. Most of us in the AI field of course understand this. The researchers wrote that popular and buzzy AI models "face a complete accuracy collapse beyond certain complexities," especially with things they've never seen before.

In a nutshell, they presented models from OpenAI, Anthropic, and DeepSeek with new and complex puzzle games and found their reasoning ability "increases with problem complexity up to a point, then declines." A pattern matching system that predicts the next token is of course not adaptable to

matching system that predicts the next token is of course not adaptable to novelty and doesn't remotely function as human intelligence agile in problem solving and capable of self-learning would.

Apple rightfully questions and argue that the existing approach to benchmarking "often suffers from data contamination and does not provide insights into the reasoning traces' structure and quality."

The debate mostly took a familiar pattern of defending Generative AI's hype trend with analysts saying it could be Apple's way of doing [corporate damage control](#). Others tried to say that novel puzzles aren't the best way to test reasoning models or these more advanced models by the likes of OpenAI, Claude, Google and DeepSeek. By using "controllable puzzle environments," the Apple team estimated the AI models' ability to "think" — and made a seemingly damning discovery.

Thanks to a "counter-intuitive scaling limit," the AIs' reasoning abilities "declines despite having an adequate token budget."

Let's finally get to our piece for today: by [Tobias Mark Jensen](#)

Futuristic Lawyer

Exploring the intersection of human rights & IT.



Futuristic Lawyer

Human Rights & IT

By Tobias Mark Jensen

The Illusion of Thinking

Apple held its annual World Wide Developers Conference (WWDC25) last Monday without any major announcements on AI. Sure, [some cool new AI features](#) were announced for the Apple aficionados but nothing we haven't

features were announced for the Apple AI-enabled devices but nothing we haven't seen elsewhere or couldn't expect. Siri is not turning into an emotionally attuned "super assistant" like [OpenAI's ChatGPT](#) and Apple is not throwing in the hat to compete with Google and Microsoft as an "AI-first company".

Apple brands itself on luxury consumer hardware, attention to detail, and high standards in aesthetics, privacy and data security. Unlike other companies that are willing to go head-first into AI like bulls in a China shop, Apple can't afford to take this route. Mistakes such as the dysfunctional AI-generated summaries of news [which Apple was forced to roll back this year](#) are much more costly for Apple, than it would be for say Meta or Google, due to its trusted brand value.

Still, Apple's bundle of announcements at WWDC25 [widely elicited a lukewarm response](#). People have come to expect only extraordinary things from Apple, but now it is sitting cautiously on the sideline while its competitors are pushing the boundaries of a new paradigm with AI. There is, however, also a chance that its passive second-hand approach to AI can pay off.

In my view, Apple's most important AI contribution came a few days prior to WWDC25 in the shape of the paper "[The Illusion of Thinking](#)". Competitors like Google and OpenAI are building their ambitious AI efforts on the idea that models can reason and that this reasoning can scale. Only then, can we truly have helpful AI agents who understand humans and accommodate our needs. Apple's new paper throws a bucket of cold water on that assumption.

Apple's internal research team - Parshin Shojae, Iman Mirzadeh, Keivan Alizadeh, Maxwell Horton, Samy Bengio and Mehrdad Farajtabar - argues why so-called Large Reasoning Models (LRMs), including OpenAI o1 and o3, DeepSeek-R1, Claude 3.7 Sonnet Thinking, and Gemini Thinking, cannot reason. It only seems like they can. I will summarize their findings in the remainder of this post.

First, we have to understand why LRMs are different from Large Language Models (LLMs).

LRMs are trained with an emphasis on logical reasoning tasks that require

step-by-step thinking. Typically, LRMs generate so-called chain-of-thought outputs in this process –kind of intermediate reasoning steps in the LRMs' "thought processes" – which they then self-verify to produce more accurate and well-reasoned responses. Users are typically able to see the model's chain-of-thought outputs but not always the underlying "thinking tokens" that go before the reasoning steps that are visible to users. For example, OpenAI's reasoning models o1 and o3 do not show users the thinking tokens (which is also why they only play a minor role in Apple's paper).

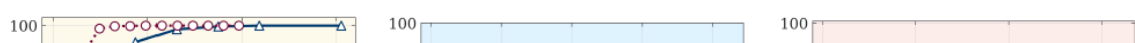
LRMs' chain-of-thought processes can take anywhere between a few seconds to a few minutes to respond, or theoretically longer. Due to the increased “thinking time”, the compute cost is also higher which makes LRMs significantly more expensive to use than LLMs (DeepSeek managed to bring these costs down drastically [but that is a story for another day](#)).

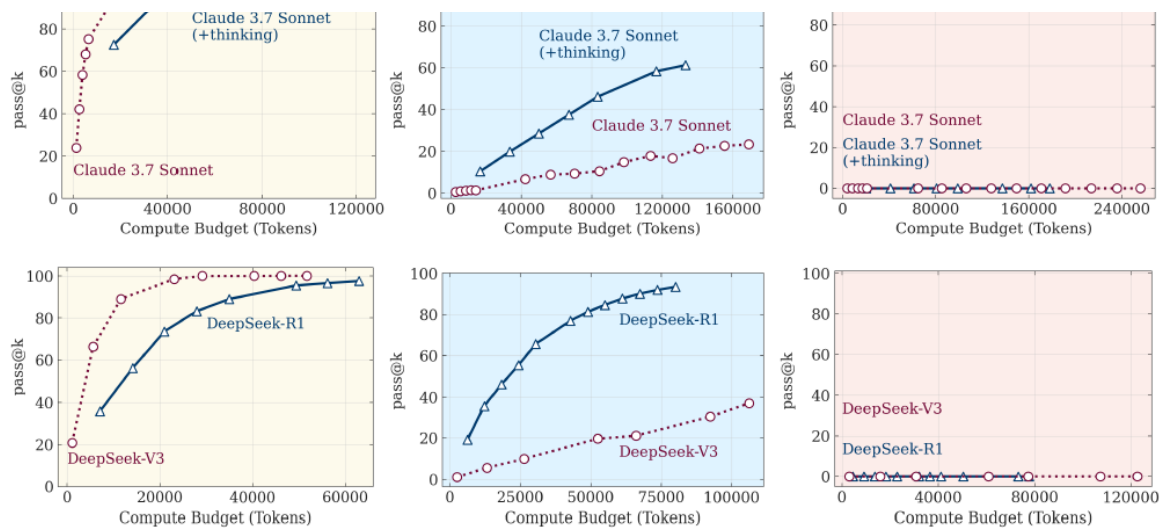
The common wisdom is that the longer LRMs take to generate a response, the more accurate and well-reasoned the answer will be. Theoretically, this means that a user who is willing to spend a lot of computing costs and a lot of time - say hours, days, and weeks - could obtain an exceptionally good response to a very hard problem. As we shall see very soon, Apple's paper disproves that idea completely.

Now, to the first experiment.

The researcher at Apple tested two LRMs – Claude 3.7 Sonnet Thinking and DeepSeek-R1 - against two LLMs without “reasoning abilities” – Claude 3.7 and DeepSeek-V3. The four models were tested across four text versions of classical puzzle games - Tower of Hanoi, Checker Jumping, River Crossing, and Blocks World. Each of these puzzles requires logical reasoning skills to solve and they can be adjusted in complexity. Apple's researchers tested the four models on “three regimes of complexity”.

The results are surprising and depicted in the illustration below (low complexity: left, medium complexity: middle, and high complexity: right)



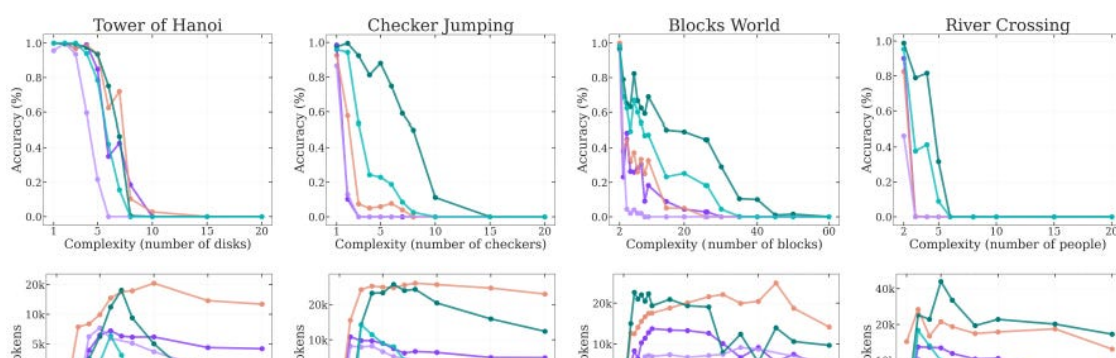


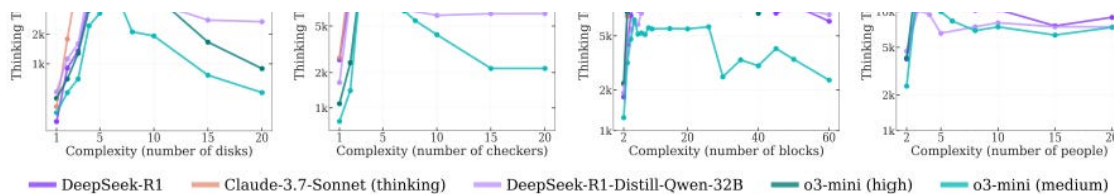
- In the first regime with a low problem complexity, the LLMs could perform comparably, or even better than LRMs.
- In the second regime with a medium problem complexity, the LRMs gained the advantage over LLMs.
- In the third regime with a high problem complexity, the performance of both kinds of models collapsed to zero.

The results of the third regime were the most vital part of the experiment and led the researchers to make the following conclusion:

“frontier LRMs face a complete accuracy collapse beyond certain complexities”.

In a second experiment, the researchers tested how five LRMs - o3-mini (medium and high configurations) DeepSeek-R1, DeepSeek-R1-Qwen-32B, and Claude-3.7- responded to increasing problem complexity. The results are depicted below.





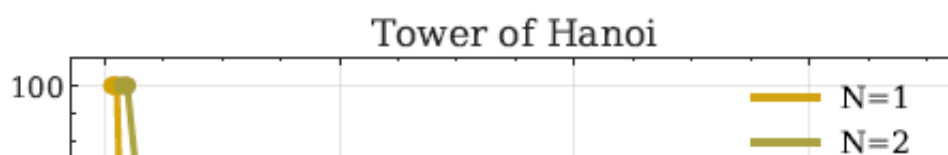
Again, we can see that all five LRMs exhibit the same pattern: as the complexity of the task increases, the accuracy progressively declines until reaching complete collapse (zero accuracy).

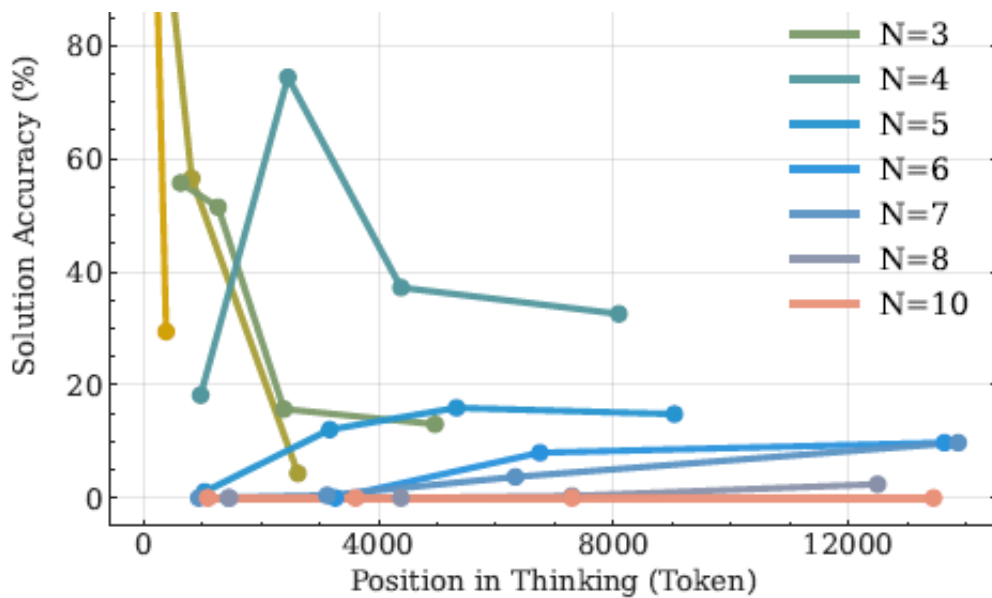
The researchers also observed that the LRMs increased their thinking tokens proportionally with problem complexity, but only initially. After approaching a certain threshold —which closely corresponded to their accuracy collapse point— the models counterintuitively began to *reduce* their reasoning effort despite increasing problem difficulty. Apparently, this phenomenon was most pronounced in o3-mini and less severe in the Claude-3.7-Sonnet-Thinking.

The results of the second experiment support that LRMs are suffering from a fundamental scaling limitation – contrary to claims by OpenAI and many others.

In a third and final experiment, the researchers studied the reasoning process of Claude-3.7-Sonnet-Thinking alone to understand the implications of their research on a more granular level.

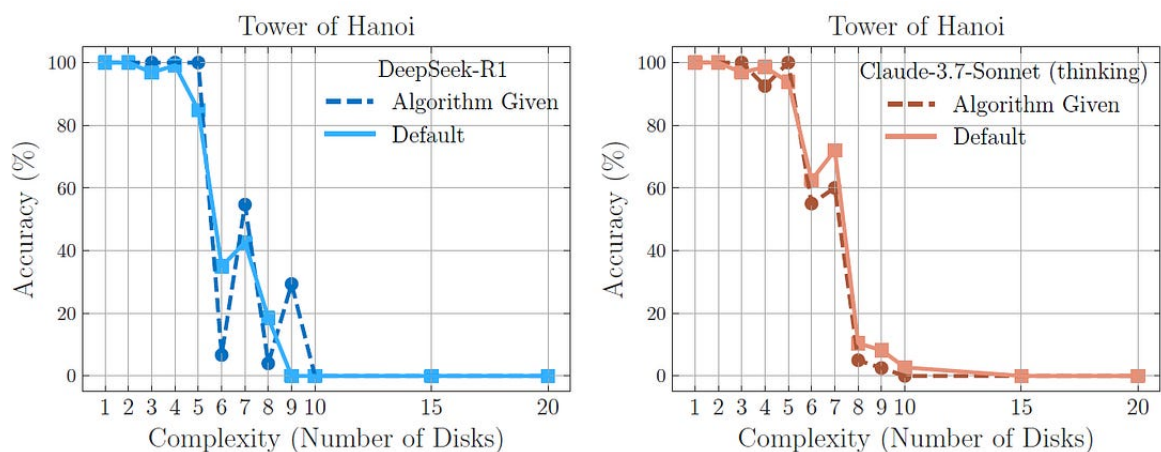
For simpler problems, Claude-3.7-often found the correct solution early in its thinking process but then continued to explore other, incorrect solutions. This phenomenon is called "overthinking" in the literature. As problems become more complex, the trend reverses – Claude-3.7 first explores incorrect solutions but later in its thinking process arrives at the correct ones. Then, when approaching problems with a still higher complexity, it fails to generate any correct solutions whatsoever.





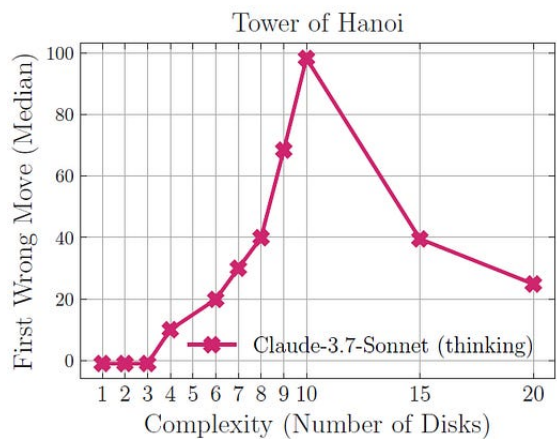
Toward the end of the paper, Apple's researchers make two very strange and noteworthy observations.

Firstly, when they provide Claude-3.7-Sonnet-Thinking and DeepSeek-R1 with the correct steps to take to solve problems with high complexity, they still fail to come up with a correct solution and the collapse in reasoning occurs at roughly the same point. This indicates that much more research is still needed to understand how these models really work.

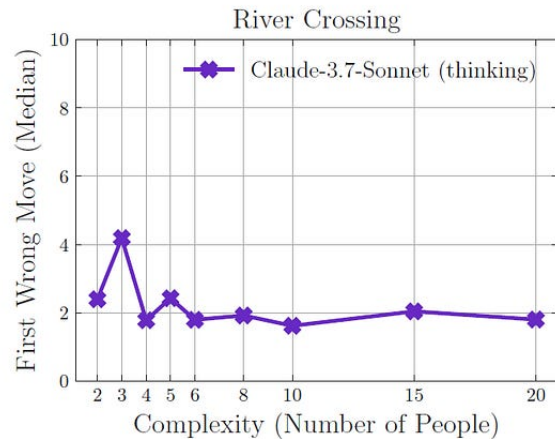


Secondly, when the researchers investigated the first failure move of the Claude-3.7-Sonnet-Thinking, it revealed very different behavior on the Tower of Hanoi and the River Crossing problem, respectively. In the Tower of Hanoi puzzle, the first error occurred around move 100 (N=10), whereas in the

River Crossing puzzle, the first error occurred already at move 4. According to the researchers, this is a clear sign of data contamination. The Tower of Hanoi puzzle must have been frequently encountered or memorized during the model's training process.



(c)



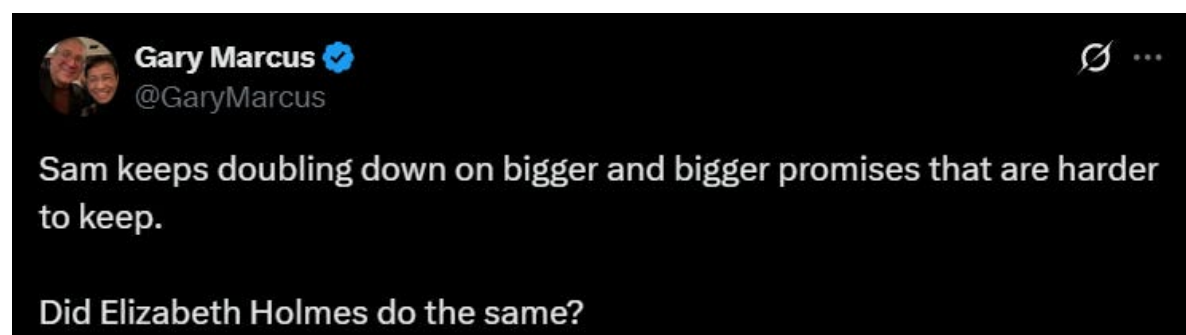
(d)

The research by Apple's team I have accounted for here is of course narrow in scope, as it's limited to four puzzles. We can't say with any confidence how well the results translate to the application of LRMs in the real world.

Nonetheless, I personally think this paper is very damning to the bull case and hype around LRMs. Indeed, I am inclined to say that Apple is making the right call by taking its cautious, sideline approach to AI.

Read more Reactions to the Apple Paper

A lot of the fame of Gary Marcus in current times appears because he is a critic of Sam Altman:

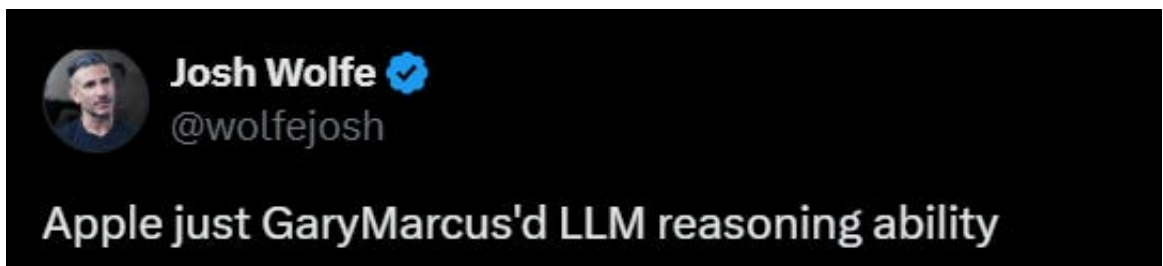


- **Critique of Pure Reasoning Models** by Harry Law
- **A knockout blow for LLMs?**
- Let's Talk THAT Apple AI Paper—**Here's the Real Takeaway the Internet is Ignoring** by Nate
- **The illusion of "The Illusion of Thinking"**

A perfect example of how academic papers in AI have degenerated into PR is the following example:

[Read Paper](#)

- Alberto also took this approach: **The Illusion of 'The Illusion of Thinking'**
- **Seven replies to the viral Apple reasoning paper – and why they fall short**



The day Gary Marcus became a verb indeed. **Anthropic in recent months** have become as aggressive as OpenAI or Google DeepMind in their Comms/PR/GTM strategy, which includes exaggerating the capabilities of their models, future capabilities of AI on humanity and the variations of the RSI pitch regarding Claude Code. This includes the PR-ification of academic papers and “studies”. **Critiques of Apple’s paper** were plentiful. Researchers from Anthropic and partner institutions have introduced **Internal Coherence Maximization (ICM)**, a method that fine-tunes language models using only their own outputs rather than human supervision, relying on internal consistency and logical coherence.

What was striking to me among AI Newsletters and my peers, is how few took Apple’s side along lines of common sense every AI research knows all

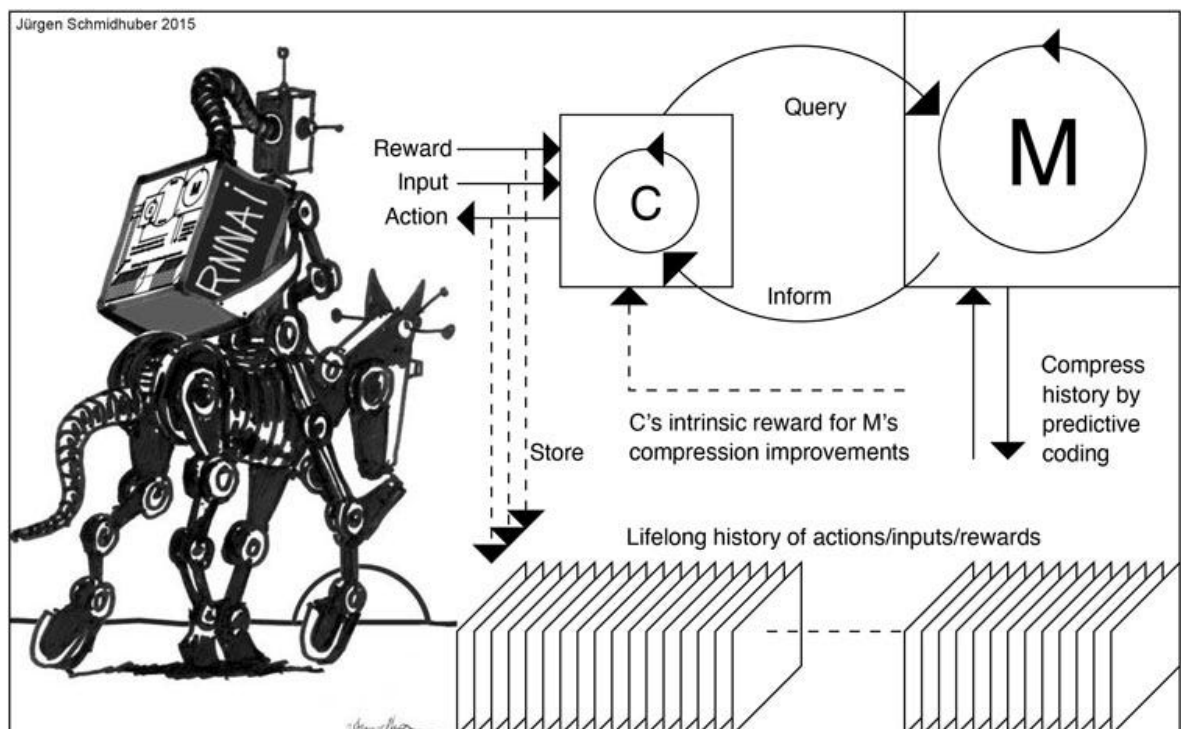
look Apple's side along lines of common sense every AI research knows all too well. When there are financial incentives to hype the narrative how does the truth bend and how difficult is it to admit the limitations of LLMs?

According to a [March, 2025 Survey](#) most AI researchers don't believe current architectures and systems can approach anything nearing real reasoning or AGI. The survey of 475 AI researchers by the Association for the Advancement of Artificial Intelligence (AAAI) conducted as part of its panel on the future of AI research found that "[t]he majority of **respondents (76%)** assert that 'scaling up current AI approaches' to yield AGI is 'unlikely' or 'very unlikely' to succeed.

For me it's just a pity that Apple's paper got more attention than the RSI papers that have much greater potential implications for what Generative AI companies will try to do with agents to get better at tasks. RSI in Agentic AI could be possible along some narrow definitions of task performance improve in automating repetitive digital or tedious tasks.

Learning it Learn

[Read the Blog](#)



POLL

Is Generative AI the path to recursive self-improving agents and Meta reasoning that transcends limitations of token prediction and hallucinations? Can AI learn to Learn without limits?

Yes

No



A guest post by

Tobias Mark Jensen

Tech writer with a legal background

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